

SHOULD I BUILD OR SHOULD I BUY?...

BUILDING WILL GIVE TROUBLE, BUYING MAKES IT DOUBLE...

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Abstract – Should I build or should I buy?... If I build there will be trouble, if I buy there will be double... When an institution is looking into starting up their own Digital Preservation System, one of the first questions will be: should we build it ourselves or buy something from the market? Both of these options have, at first glance at least, advantages and disadvantages. The National Library of Norway (NLN) and the National Library of the Netherlands (KBNL) discussed this as part of an internal workshop where the NLN presented how they built an on-premise system, and the KBNL presented the implementation of their purchased system. These different viewpoints and approaches in establishing a digital preservation system might help others to better understand whether they should build or buy.

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I. INTRODUCTION

During iPRES2024, Inge Hofsink of the National Library of the Netherlands (KBNL) presented a paper in which she shared her personal reflections about the impact of organizational context and culture on thirty years of digital preservation at her institution. [1]

This paper led to a workshop meeting between the National Library of Norway (NLN) and KBNL. [2] Two of the main topics were organizational challenges regarding preservation, and digital

preservation systems (DPS). The latter of these will be the topic of this paper.

NLN has chosen to build their own DPS system[3], while KBNL has chosen to buy a system. In this paper we compare the two different approaches towards acquiring and implementing a DPS. This includes both the reasoning behind the choices, as well as describing context in which they were made.

The question of building or buying has a large effect on what an institution can and can't do with a system. Buying forces one to conform to an in-place framework, while building gives a lot more freedom to create something tailor-made but it also results in more time in development and testing. Or does it? Do these statements really hold true, and if not, why not?

II. THE START: WHERE TO GO FROM HERE

The old systems of both the NLN and KBNL have been running for multiple years, sometimes even longer, and are in need of replacement. So the question both institutions were asking: now what? This led to both discussions within the organizations and a list of requirements used to choose the way forward.

The NLN primarily based their digital preservation on safe storage using ORACLE HSM (SAM-FS) as bit-repository. One third of the data had

no checksums and archived data was packed and stored in multiple ways. There was no overall information about the content and use of it. ORACLE HSM was approaching EndOfLife and needed to be replaced. This opened an opportunity to establish a more comprehensive Preservation solution making it possible to work more systematically with digital preservation. NLN also decided to establish an interdisciplinary digital preservation team consisting of IT- and metadata specialists. In the search for a new DPS it was important that the solution could cater for large data volumes (15+ PetaByte) and expected growth. It must support automated processes and built on standardized and open solutions. In addition to that, the NLN has divided its digital objects into separate preservation and access solutions. After surveying the market and evaluating alternative solutions, the NLN found that existing systems presented several challenges. These included limitations in handling large data volumes, licensing models that were often volume-based, incompatibility with the NLN platform environment, or functionality that was not needed or requested. Instead, the NLN drew inspiration from approaches taken by comparable institutions and decided to build a DPS in 2022.

The KBNL started this process back in 2017 with among other things the creation of a dedicated preservation department. It quickly became apparent that there was a preference to buy a system over building a new one. The latter had been done for previous system, with less than successful results. One of the main draws of a purchased system, and a main requirement, was the wish to have a user community that could be consulted with regarding preservation. Another important requirement was the ability to handle large to very large amounts, at least 1 billion files, of data. This requirement has been tested extensively by both KBNL and Ex Libris; the latter even created a specialized environment [4] for this. In the end, KBNL bought the Ex Libris Rosetta system and started implementing this in the years following 2020. The teams behind the buying and testing of the system consisted of preservation specialists, IT-specialists, architects and information specialists.

As can be seen, both institutions chose a different approach to get a new system. There is of course no right or wrong. What is interesting, and which was noticed during the workshop in November 2025, was that while the implementation of the two systems was different, there was a lot of

overlap in thinking behind the different aspects. Aspects like how to deal with ingest and preservation itself. That's not saying that everything is similar, since there are some major differences in both institutions scope and certain choices.

III. IMPLEMENTATION

Based on the experiences of surveying the market, the NLN chose to create their own solution. This coincided with the start-up of a preservation project that was initiated by the ministry of Culture and Equality. This ensured necessary commitment from the NLN's management. They also had the necessary IT resources available, in terms of personnel and infrastructure. The NLN chose to use a LEAN [5] approach to developing a new preservation system.

The choice made by KBNL for a bought system was influenced a lot by the failure to complete the creation of a self-build system.

The KBNL had a less than stellar experience with building a system. Together with other requirements, like having a community related to the system, resulted in the choice to buy a DPS. The tendering process for the system took a lot of time in and by itself, taking about two years from start to finish and many hours of work for multiple specialists. But this was not the end. When Rosetta was chosen as the DPS the detailed work started. This included looking in depth at the data model and making choices regarding the information KBNL wanted to record, how to record it, and selecting a working and correct metadata implementation within the data model. Again, this was not something that was done overnight, taking multiple years. While this was some of the main work to be done to be able to map data towards this data model, it was only the first of two main steps. The next would be to have something that would be able to transform any incoming format to the system required format. While not part of this paper, this was done via a self-build pre-ingest tool. This tool is mentioned to clarify that there still is a need for at least some custom work, even if an institution buys a system from the market. KBNL's tool was built via the Agile manifesto and with a team of specialists and IT developers.

While buying or building is the first step, the main goal of a DPS is to have a system to preserve materials. The implementation requires all parties to think about use-cases, best practices and use plain

common sense to be able to have a way of working that works for them.

IV. DATA MODEL

While the data models for both systems differ, there are also some commonalities. The NLN decided on a model based around e-ARK [6], where the KBNL uses Rosetta's METS [7] implementation. Both are METS or METS derivatives. Furthermore, where the NLN's have an extensive use of links to external files, the Rosetta METS contains almost all the information in one XML file.

One big difference between the KBNL solution and the NLN DPS is that the NLN DPS doesn't require any standards for descriptive metadata. Suppliers can include metadata as part of the delivered package in any form they choose. In addition to that, the DPS Submission API allows suppliers to deliver a minimum set of metadata based on DublinCore [8]. This metadata is first and foremost for use within the DPS. Every object in the DPS has a unique identifier which is expected to be stored in the consumers cataloguing system. This is used to retrieve the item in question.

The KBNL on the other hand worked out a data model implementation for descriptive metadata, also around DublinCore. KBNL's choice for this implementation was supported by arguments stating that descriptive metadata is needed to identify and/or recognize materials. Identifiers, persistent or not, were another point that was discussed and later implemented. Both the identifiers from the publication itself, as well as persistent KBNL generated ones, were implemented.

For the NLN DPS, the suppliers can include preservation events as part of the delivered package in any form they choose. This information is only relevant for the supplier and has no requirements regarding form or content. It will just be stored and not used within the DPS. In addition to that, the Submission API allows for a more standardized approach where suppliers can deliver a certain set of preservation metadata events based on PREMIS. These metadata will be validated and stored together with preservation events produced within the DPS.

For the KBNL there were multiple internal working sessions to discuss events. These sessions dealt with both the legacy events, events from the previous two DPS, as well as events related to new

material that would be ingested after passing through our pre-ingest system. KBNL is restricted to the Rosetta data model, which means that, while it's PREMIS based, it's not PREMIS and this was something that needed to be taken into account. In the end a choice was made to make use of the Rosetta formatting for events but customized with KBNL eventIDs combined with mainly eventTypes as described by loc.gov. [9]

V. REFLECTIONS

But returning to the main point of this paper: what are the advantages and disadvantages of buying or building a Digital Preservation System? It is difficult to determine whether many of the points discussed earlier should be considered advantages or disadvantages for either option. For example, building a system allows for a high degree of flexibility, but this flexibility can also complicate implementation. Clear and fixed scopes are necessary, since without them the product risks becoming unfocused and inconsistent. In such cases, the result may be a system that attempts to address many needs without excelling in any of them.

A similar issue can occur when purchasing a system. Although institutions must work within the framework of a predefined model with limited possibilities for customization, the resulting system may still fail to fully meet institutional requirements. For example, institutions may need to develop workarounds to record certain types of information, such as legacy preservation events or highly granular descriptive metadata.

Another important aspect of the buy-versus-build decision is development and implementation time. As expected, building a system requires a team of dedicated professionals including IT, working on the project continuously. Buying a system may initially appear easier, but it still requires a team to evaluate the new system and ensure that both the software and the existing data fit within the institution's workflows. While this team may not require extensive programming expertise, significant discussion and testing are necessary to determine how the system will be implemented and how it handles different processes.

Experiences at KBNL showed that the Rosetta system sometimes behaved differently from what was initially expected. Although this was not detrimental, it was nonetheless an unexpected characteristic that required adjustments. It was also

concluded that the overall process—from initial planning to implementation—took considerable time. Even after this extended period, the implementation of the system at KBNL remains ongoing, illustrating how long such projects can take.

Finally, purchasing a system means that an institution commits to a contract with various requirements. This creates a relatively fixed strain on the budget, and public procurement regulations require systems to be re-tendered periodically, which introduces additional costs. Furthermore, institutions become dependent on the financial stability of the supplier. If the supplier were to go out of business, significant risks could arise. Both the need for re-tendering and this dependency highlight the importance of clearly defined exit strategies. Without such agreements, there is a risk that institutional data may be difficult or costly to access and migrate to a replacement system.

VI. CONCLUSION

Building a system offers the advantage of maintaining full control over how the system evolves, as well as greater control over development and operational costs. However, it also involves a higher risk of failure, and institutions must manage these risks largely on their own. These include the dependency and access to key personnel, scope creep and risk of technical debt.

For most institutions, purchasing a system therefore appears to be the safer option. If an institution chooses to build its own system, it is essential to ensure that sufficient IT and infrastructure resources are available for a long time to sustain the project. It is also advisable to adopt a LEAN/agile approach, implementing functionality incrementally over time.

However, purchasing a system is not without its own challenges. A formal tendering process is often required, and due to European regulations and institutional procedures, this process can be lengthy. While purchasing a system may sometimes be cheaper than developing one internally, selecting a commercial product commits the institution to a multi-year contract with financial obligations that may be difficult to adjust if budgets change. For institutions that rely on government funding, significant policy changes can therefore create additional financial pressure.

Furthermore, institutions must prepare and agree upon clear exit strategies. These are necessary both when a new tender results in the selection of

another system and when a supplier faces financial difficulties or bankruptcy. Depending on the contractual arrangements and technical implementation, institutions may risk additional costs or difficulties in retrieving their own data.

Ultimately, the question remains: should an institution build or buy? As this paper has shown, both options involve advantages as well as challenges. The final decision depends largely on the resources, expertise, and risk tolerance of the institution. Some degree of difficulty will almost certainly arise regardless of the chosen approach — but whether that difficulty becomes “double trouble” remains an open question.

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